# Completing the Components of Project Charter

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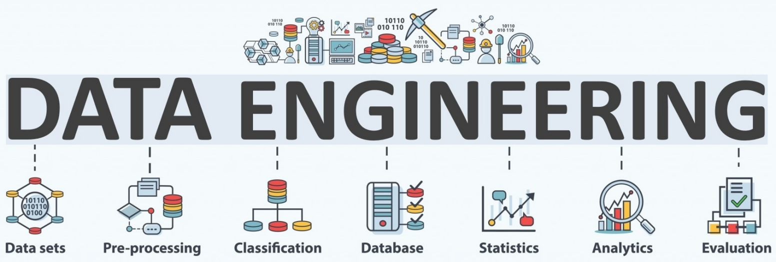
**Data engineers get data and provide value from the data.**

- A data engineer is expected to know and understand how to use a small handful of powerful and technologies to create a data solution.
- Utilizing these technologies often requires a sophisticated understanding of software engineering, networking, distributed computing, storage, or other low-level details.
- Technical responsibilities includes: generation, storage, ingestion, transformation, serving.



# Project Charter Context & Value

- What is a project charter?

  **Short** document that officially starts a project.

  **Defines** what the project aims to achieve and who is involved, helping align everyone before the project starts.

- Why it's critical in data engineering projects?

  Alignment
  Scope
  Expectations

- How it sets the stage for successful execution of the capstone?

  It creates a clear roadmap and ensures everyone **understands** what needs to be done, by whom, and why.

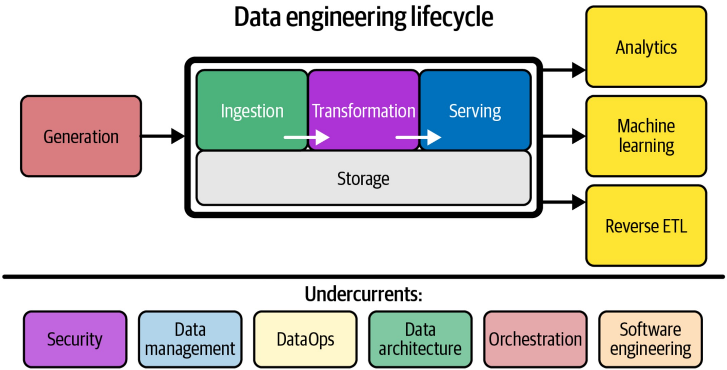
- Next, core components of a project charter ...

# 1. Project Overview & Objectives

- Brief description of the capstone.
- SMART goals
  (e.g., process X TB of data, improve ETL runtime by 30%).

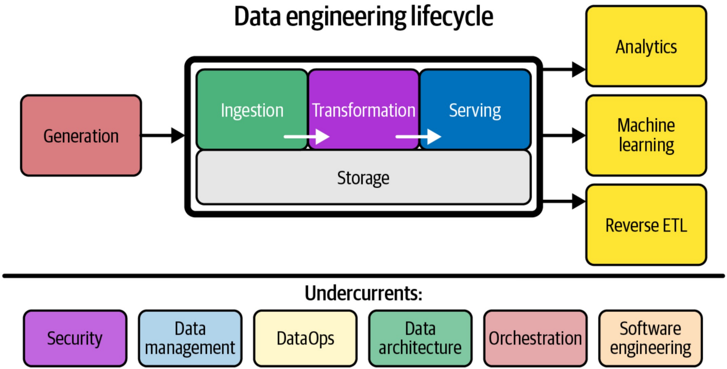
| Letter | Meaning | Explanation |
| --- | --- | --- |
| S | Specific | Clearly define what you want to achieve. |
| M | Measurable | Ensure progress can be tracked or quantified. |
| A | Achievable | The goal should be realistic and attainable. |
| R | Relevant | Should align with the project or overall objectives. |
| T | Time-bound | Set a deadline or time frame to complete the goal. |

SMART Framework for Goal Setting




**Generation:** Data is created or collected from various sources such as applications, sensors, logs, or external systems.

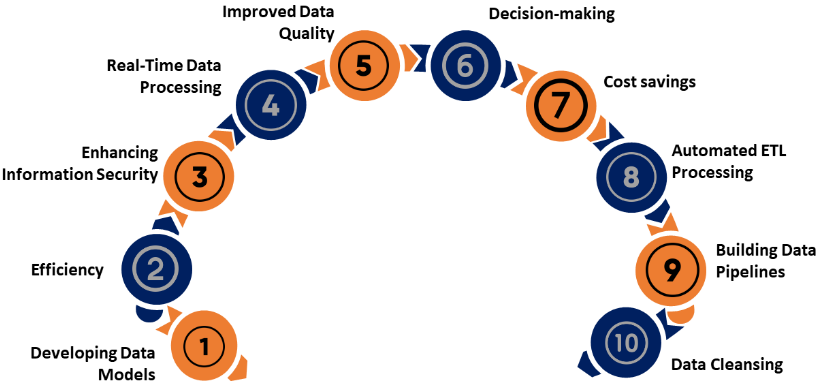
**Ingestion:** Data engineers build pipelines to bring data into the system—streaming or batch. This is where extraction and initial loading occur.




**Transformation:** Data is cleaned, standardized, enriched, and modeled. This stage ensures the data is usable and trustworthy.

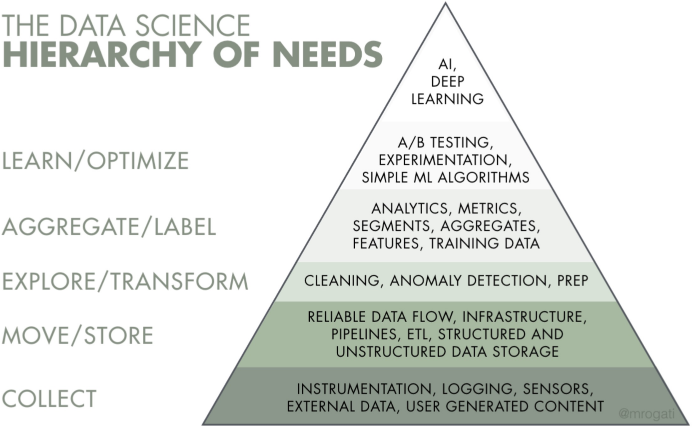
**Serving:** Processed data is made available for consumers. This could be via databases, APIs, or data warehouses/lakes.

# Advantages of Using Data Engineering Tools



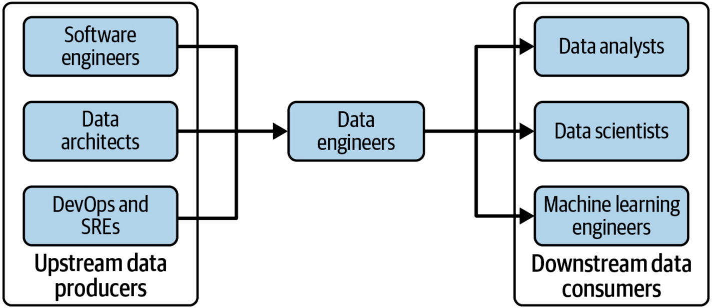
# 2. Scope Definition

- In-scope vs out-of-scope components.
- Data sources, pipelines, and deliverables (dashboards, ML feature store, documentation).



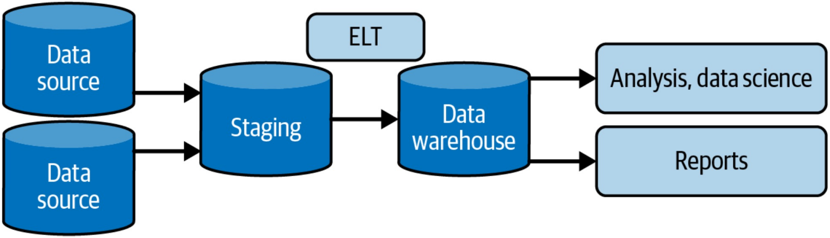
# 3. Stakeholders & Roles

- Project sponsor
- Technical lead
- Data engineer(s)
- Advisor / mentor
- End users



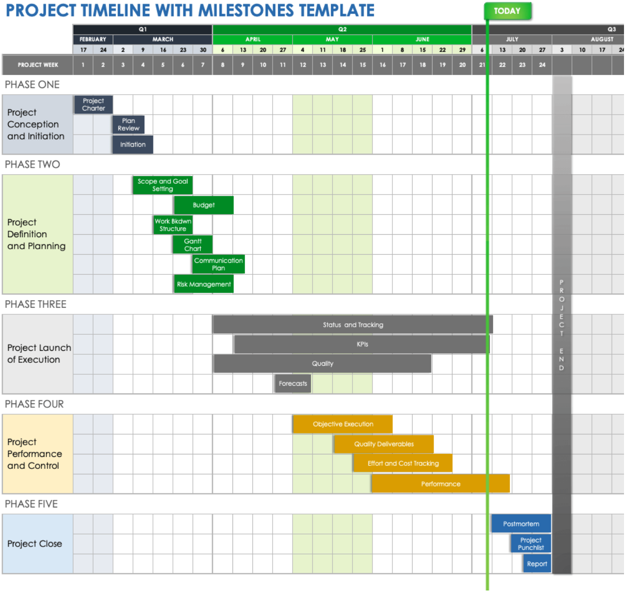
# 4. Project Deliverables

- ETL pipeline prototype
- Data model / schema
- Storage solution (e.g., warehouse, lake)
- Performance benchmarks
- Final presentation/document



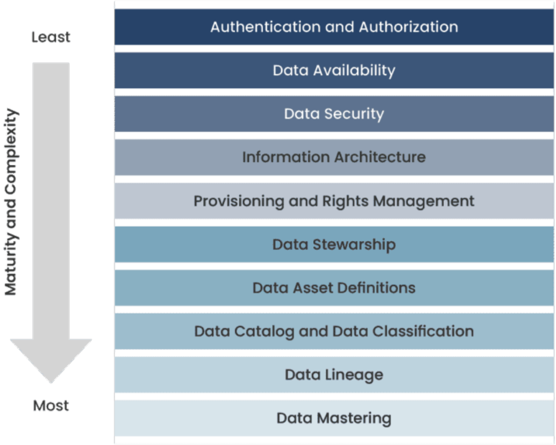
# 5. High-Level Timeline / Milestones

**Kickoff → Requirements → Design → Development → Testing → Deployment**



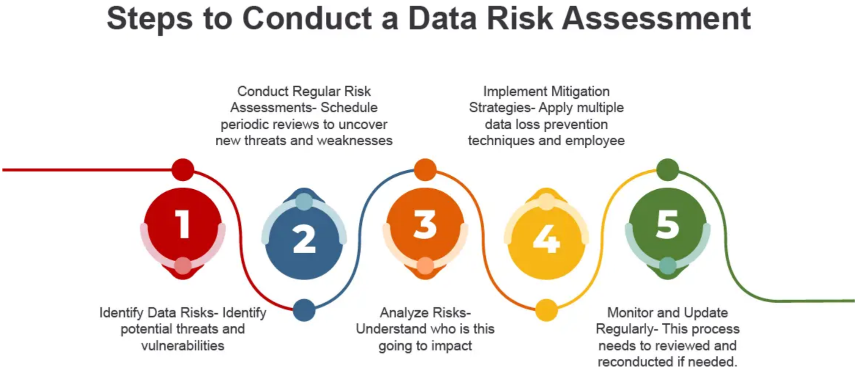
# 6. Assumptions & Constraints

- Data availability
- Data standards
- Tools, time and resource limits



# 7. Risks & Mitigation

- Data quality issues $\rightarrow$ define cleanup strategy
- Performance limits $\rightarrow$ plan scaling options
- Dependency on external APIs $\rightarrow$ backup datasets



# Conclusion

**"A clear charter leads to a clear path."**

- Clarity and alignment
- Validate goals using data feasibility
- Ensure risks & constraints reflect real limitations